Lab 3 Report: Uninformed Search

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Objective

- To understand different uninformed search algorithms.
- To implement BFS, DFS, DLS, IDDFS and UCS in Python.
- To find a path from the starting node to the goal node.

Tasks Performed

Task 1: BFS and DFS

- Implemented Breadth First Search (BFS).
- Implemented Depth First Search (DFS).
- Tested both algorithms on a given graph.
- Checked the visiting order and path to the goal.

Task 2: DLS and IDDFS

- Implemented Depth Limited Search (DLS).
- Tested the search with different depth limits.
- Implemented Iterative Deepening DFS (IDDFS).
- Checked how the algorithm finds the goal at different depths.

Task 3: Uniform Cost Search (UCS)

- Created a weighted graph with different path costs.
- Implemented Uniform Cost Search using a priority queue.
- Calculated the total cost of the selected path.
- Compared UCS with BFS.

Result

- All three tasks were performed in Python.
- Different search algorithms were tested successfully.
- The algorithms found the required paths.

- UCS selected the path with the lowest total cost.
- The test cases were used to check the results.

Conclusion

In this lab, I learned how different search algorithms work. I also practiced finding paths, using depth limits and calculating path costs in Python.